Coulomb's law: $F_E=k\ q_1q_2/r^2$
Coulomb's Constant: $k \approx 9 \times 10^9$ | = 1/[4* pi * ϵ_0] or permittivity of free space

Electric Field: E = k q/r^2 * r_hat (multiply this by a charge and its distance to get force w/ direction) | F = qE
Uniform electric field formed by two oppositely charged plates next to each other. Charge density = Q/A

Magnetic force can usually be ignored in most electric field problems, but dont forget abt it
A concept not in Physics 1 is finding the electric field at a point from a uniformly distributed rod or plane, which requires calculus. I think you should be able to figure it out p easily.

$$E = [kQ/L]\int_a^{a+L} dx/x^2$$

$$E = kQ/[a(a+L)]$$

Rod | a is distance from rod | L is length of rod

$$\mathbf{E = kxQ/(R^2 + x^2)^{3/2}}$$

Line | R is radius of ring | x is distance from ring

Can do the same for a disk by integrating across rings.
Generally, E = integral dE = integral electric field equation where q is charge density and r changes

Gauss' Law | Flux (phi) is basically flow, through a surface
(flux) Phi = E * A * cos(theta)
FLUX = $q_{enclosed}/\epsilon_0$ | 4pi*k* $q_{enclosed}$
Flux in = flux out when no contained charges

For non uniform electric field: usually can simplify to flux in + flux out where one is negative

To apply, usually set EAcos(theta) = 4pi*k*q_enc to find E at a certain distance

Curved surfaces: phi = integral [E] for dA.

Rotation of Dipole (paired + , -) in uniform electric field: p (dipole moment) = qd | q charge | d dist from axis of rotation (midpoint prolly)
and torque = E * p * sin(theta) IN TOTAL? Since dipole moment is in total?

REVIEW

$E = \lambda/2\pi\epsilon_o r$	Infinitely long line of charge
$E = \sigma/2\epsilon_o$	Infinitely big plane of charge
$E = Q/(4\pi r^2\epsilon_o)$	Outside a shell or a sphere of charge
$E = 0$	Inside a shell of charge
$E = Qr/4\pi\epsilon_o R^3$	Inside a sphere of charge
$E = \rho R^2/(2r\epsilon_o)$	Outside a solid cylinder of charge
$E = \rho r/(2\epsilon_o)$	Inside a solid cylinder of charge

Unit 2:

Remember that external work done = change in mechanical energy = PE + KE
And work done in system is change in KE from PE loss
$K_1 + U_1 + W\_external = K_2 + U_2$
V IS ELECTRIC POTENTIAL AND IT IS BASICALLY A RELATIVE QUANTITY (Joule/Coulomb or Volts)

Change in **V = integral E * dx = –Δx * E** when E is constant. | **E = V/Δx** too. Change in potential is negative as you move along E.
V = kq/distance from charge. Change in electric potential is the same for + and – but PE&KE not
$\Delta U = q\Delta V$ btw

Can convert from joules to eV “electronvolts” btw, which is the amount of KE gained by accelerating electron through a space with 1 ΔV.

Equipotential surfaces are basically contour lines for electric potential. Of course, the densest areas are where ΔV is the greatest in the shortest amount of distance or where |E| is the greatest.

Electric potential at points next to uniform-charge surfaces (THIS IS TRICKY, PRACTICE)
Option 1: V = integrate dV along something we know the potential of, such as inf small charges (**V = kq/distance from charge or dV = k*dq/r**)
Option 2: integrate dV = integrating -Edr if we know E from the unit above

ELECTRIC POTENTIAL OVERVIEW

$\Delta V = -\int E \cdot ds$

- For a point charge: $V = kq/r$ (If $V=0$ when $r = \infty$)
- For a uniform field: $\Delta V = -Ed$

For a uniform ring of charge: $V = kq/(x^2 + R^2)^{1/2}$

For a uniform plate of charge: $V = 2k\sigma\pi[(x^2+R^2)^{1/2} - x]$

For a solid sphere of charge:

- If $r > R$: $V = kq/r$
- If $r < R$: $V = kq(3-(r^2/R^2))/2R$

$-dV/dr = E$ $E = kq/r^2$ $F = qE$ $\Delta U/q = \Delta V$

Electric Field on Conductors in Equilibrium (non-moving):

1. Electric Field within the conductor is 0
2. Any excess charge will be on the surface
3. Electric field from excess charges will be perpendicular to the surface
4. Charge will accumulate on the flatter areas of the surface

This is tricky. REVIEW.

Generalized Potential Energy Function: **$U = q_0V = q_0q_1k/r$ | FOR PAIRS OF CHARGES**
Potential Energy Curve: F = -dU

UNIT 3:

Capacitor

Electric field in between: $(1/\text{epsilon}_0)$*charge density
V = -Ed
C(capacitance) = Q/V = **epsilon_0 * A / d** | in F or Farads | parallel plate

Capacitance for spherical capacitor: C = **AB/[k(B-A)]** for radii B > A

- To find capacitance, use Q/V, then use dV = -E and integrate dV which is integrating -E dr which is integrating E = kQ/r^2 for 'outside a shell'

Capacitance for Isolated spherical capacitor: **C = R/k**

Calculate cylindrical capacitor similarly: **C = L* 1/[2*k*ln(B/A)]**

ENERGY IN CAPACITOR (GENERAL): ½ **C*ΔV^2 | ½ QV | ½ Q^2/C**
ENERGY IN PARALLEL PLATE: ½ **epsilon$_0$ E$_{avg}$^2**

COMBINING CAPACITORS: (Because $V_{tot} = V_1 + V_2$)

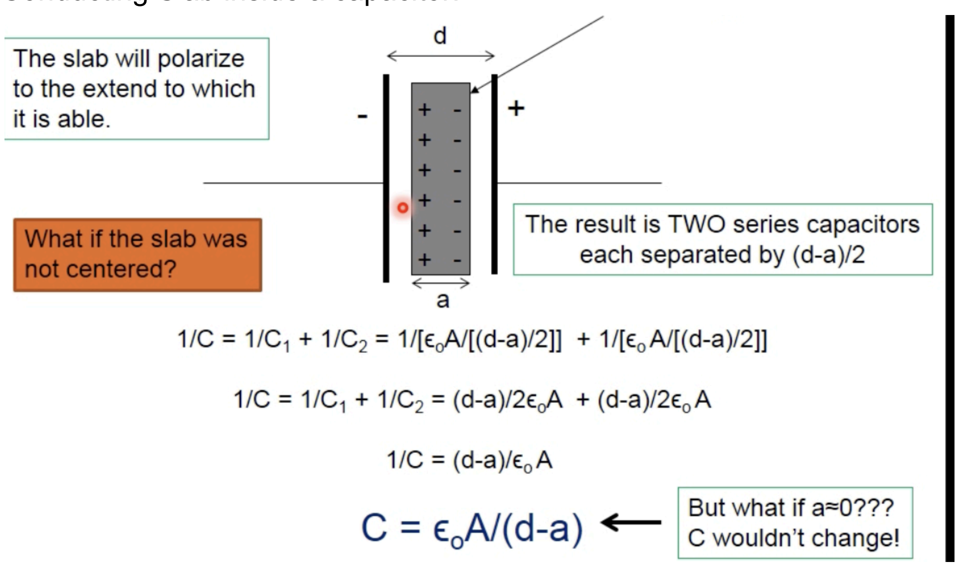
PARALLEL: $C_{EQ} = C_1 + C_2$

IN SERIES: $1/C_{EQ} = 1/C_1 + 1/C_2$

Dielectrics: Increase capacitance
Notation:
C_0 - Capacitance w/o dielectric
C - Capacitance w/ dielectric
C = kC$_0$ | k is the dielectric constant (dont confuse this 1/4pi epsilon$_0$ lol)

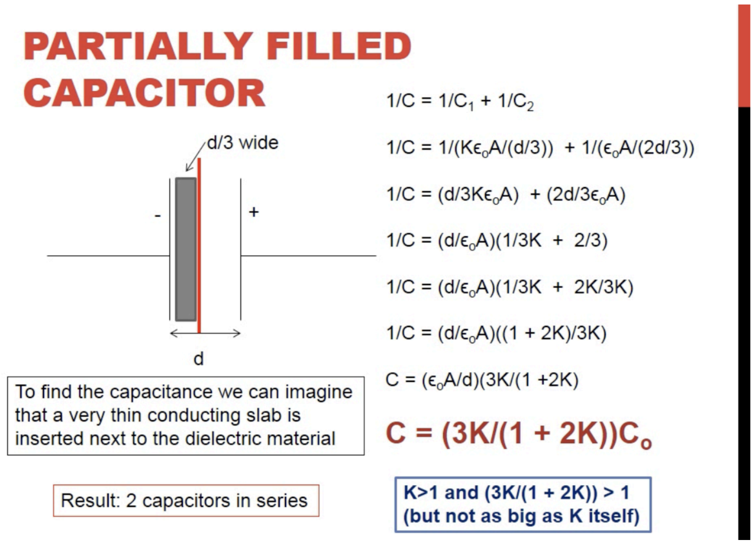
Epsilon = k*epsilon$_0$ | permittivity of the dielectric | k can represent relative permittivity (epsilon/epsilon$_0$)

Conducting Slab inside a capacitor:



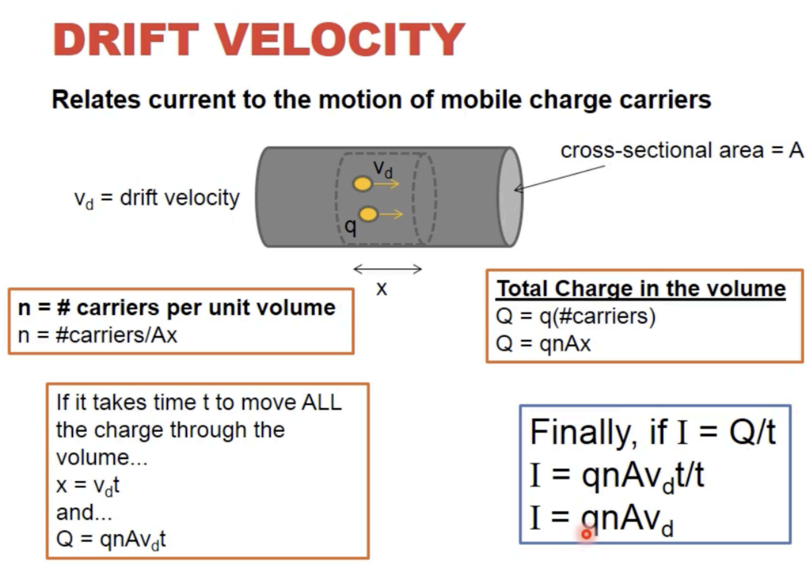
Frame of thought: Consider them series capacitors!

When the slab is not centered, no difference.



Usually dielectrics take up the whole inside of a capacitor, but if they dont, we can just use capacitors in series to solve this. One with length d/3 but multiplied capacitance by the dielectric constant, and the other length 2d/3.

Drift Velocity (v_d):



$$v_d = \frac{I}{neA}$$

q/e is the charge of an electron

Current density (J): Current is the flux of current density (vector)

$$I = \int \mathbf{J} \cdot dA$$

If J is uniform across the area, it becomes I = JA

Generalized Ohm's law: J = constant * E

Constant = sigma = **conductivity constant**. = **σ**

Also known as ohm's law: delta V=IR

R = L/(sigma A) | unit: ohms Ω

Resistivity = 1/sigma = 1/conductivity = rho = **ρ**

R = **ρ L/A | L = length | A = cross sectional area**

<u>Calculus way to find resistance</u>: integrate **ρ 1/(AREA IN TERMS OF x) dx from 0 -> L**

Electromotive Force/Device: Any device that turns non-electrical energy into electrical energy | volts

Cathode/Anode terminology is flipped when you're talking about emf devices or anything that produces energy i think

Electrical Power: P = dU/dt = q*dV/dt = I*V = V^2/R = I^2*R

Steps to finding energy used over time: Given ⅔ of I,R,V, or info about cross sectional area you can use equations for resistance, ohm's law, to calculate current, then use electrical power equation to find power (watts), multiply this by time (in hours, probably, to end up with kilowatt hours). To then find the number of joules used, you just convert from kwh to J.

Since power is correlated with cost, we can also show that higher resistance creates less power which causes less cost, which is unintutive.

CIRCUIT ANALYSIS

V=IR
ε is often the symbol used for the initial change of V at… battery?
Resistors:

- Parallel: $1/R_{EQ} = 1/R_1 + 1/R_2$
- Series: $R_{EQ} = R_1 + R_2$
- V Drop: V=IR

Capacitors:

- Parallel: $C_{EQ} = C_1 + C_2$

Series: $1/C_{EQ} = 1/C_1 + 1/C_2$
V Drop: V=Q/C

Net potential end up being 0

When branching: current is different, but $I = I_1 + I_2$. Both branches have the same drop in V.

Kirchoff's Rules:
Point rule: Current into a junction = Current out of that junction
Loop rule: Sum of potential differences is 0 for a closed circuit

RC circuit: Resistor & Capacitor Circuit w/ switch

Transient Current: ε - Q/C - IR = 0 -> V/R = I -> ε/R - (Q/C)/R = I

Once a capacitor is fully charged, current out of the battery is 0. This aligns with ε = Q/C

CHARGING CAPACITOR: $Q(t) = C*\varepsilon(1-e^{-t/RC})$
DISCHARGING CAPACITOR: $Q(t) = C*\varepsilon(e^{-t/RC})$ | Q = Cε | $q = Q(e^{-t/RC})$

Transient Current = dQ(t)/dt | Current is the derivative of charge

We often assign tau (**τ**) = RC | resistance * capacitance | $I = I_0*e^{-t/tau}$ | $Q(t) = C*\varepsilon(1-e^{t/tau})$ | I_0 = initial current

MAX ENERGY STORED BY CAPACITOR: E = ½ CV^2

Magnetism

F_B = qvBsin(theta) | theta is angle between velocity and field
B-Field unit: Tesla

RIGHT HAND RULES:

Magnetic Force: Thumb as the velocity, index as the B-Field, palm as the resultant force

Coils: Thumb as the B-Field, Fingers coiling in the direction of the coil-current.

Straight wire: Thumb as current, B-Field is circular (tangent) around the wire in the direction of fingers

Circular motion of particles in magnetic field: f = ma = mv^2/r = qvB | r = mv/qB

Helical Paths: When velocity and B-Field are not perpendicular, creates helical spring-like path along the b-field (similar to stationary circles, but now there is a small component of velocity that doesnt change along the B-field). Angular velocity w = v/r = rqB/mr | Time for a loop = 2pi/w

Biot-Savart Law:
$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I\, d\vec{\ell} \times \hat{r}}{r^2}$$

Used to find B-field distance away from current

AMPERES LAW (Straight Wire):
$$B = \frac{\mu_0 I}{2\pi r}$$

(Coils):
$$B = \mu_0 n I$$

n = turns per meter